CMPE137
Final Report

Group 02

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Date: 5/19/25

1. Idea

1.1. Description

Our project focuses on developing a marketplace app specifically for San Jose State University students. The goal is to provide students a dedicated platform to buy, sell, and trade items within the university community. Unlike Facebook Marketplace or Craigslist, this is a campus-only ecosystem for secure, student-to-student exchanges.

1.2. Goals/Objectives

There are several goals/objectives for this app, with those being to enable students to list items for sale and browse the listings of other students, ensure that the platform is restricted to only SJSU students to ensure privacy and security, simplify the buying/selling experience with intuitive UI, and to incorporate location-based filtering to enhance relevance for specific users. The constraints are an implementation of the following in one form or another: a functional prototype and design, privacy and security, connectivity (via Wi-FI, Bluetooth, or NFC), Location, Data Management + Exchange, and Notifications.

For listing items for sale and browsing the listings of other students, a messaging system will be implemented to allow for easy communication between clients. Listings will also include the general area where the seller resides (specifically just the city and zipcode) if voluntarily entered. This data and other personal information will be encrypted and stored using *Firebase Cloud Storage*, which encrypts data using Hypertext transfer protocol secure (HTTPS). Firebase Cloud Messaging will also be used to handle any notifications for the app. Limiting the app to SJSU students only will be done via the implementation of the google sign in/signup for emails ending in @sjsu.edu only. This ensures that only active SJSU students will be able to use the service. The UI will be based of a combination of the following sources: SJSU Sammy App, Facebook Marketplace, and Craigslist.

1.3. User Scenarios & Device Kit

With the target audience being SJSU students, this platform becomes useful for those living on or near campus trying to get rid of unused items. For example, students who have textbooks for previous classes they've taken, students moving out trying to sell their used furniture, or even students trying to sell their vehicle. As for the Device Kit, the app will be developed with Android smartphones running Android 8.0+ in mind.

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2. Work Developed

2.1. Design & Prototype

2.1.1. Description & Analysis

The first prototypes for the application were drafted using Figma. Here, we took inspiration from Facebook Marketplace, while keeping the general color scheme and taskbar from the SJSU Sammy app for consistency. This was initially prototyped as an addition to the existing Sammy App, but later changed to be standalone.

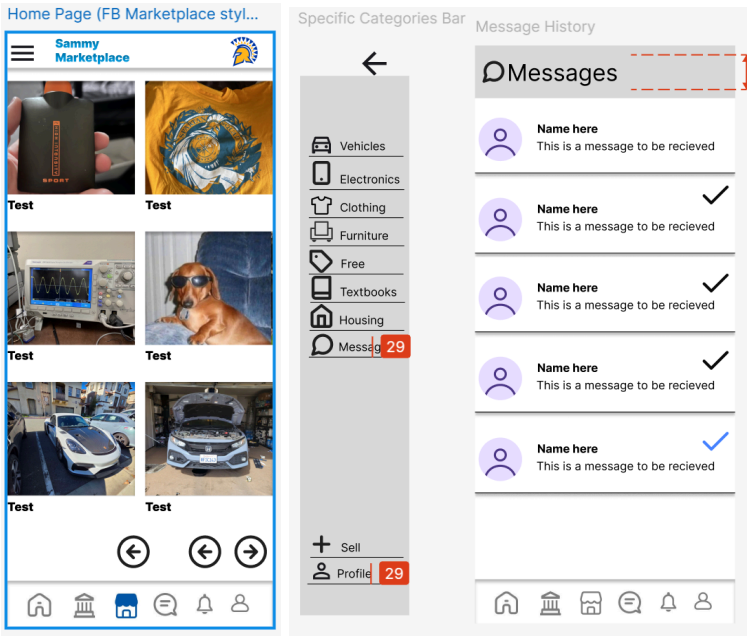


Figure 2.1 & 2.2 - First Prototype for the homepage of the SJSU Marketplace App, task bar, and message inbox

Recommendations for improvement include changing the sizes of certain elements to accommodate smaller devices, switching to a scroll view instead of buttons to change pages, and to be more consistent with design choices. An improved version of this design was intended to be fairly final, but a couple problems with converting the layouts from figma to kotlin resulted in various changes in the layout to try and get something to work. Different layouts were tried, but each had different problems varying from incorrect scaling to devices to elements not appearing all together. The version present now does not resemble this prototype and is very much a placeholder, but it retains a lot of the main layout's elements.

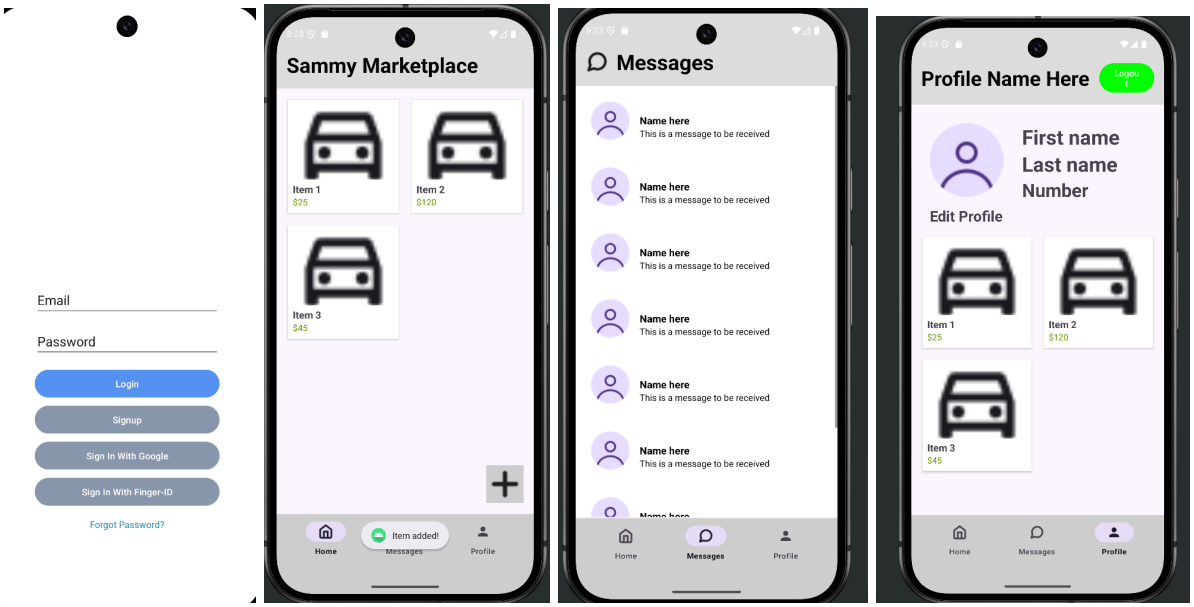


Figure 2.3-2.6- Login, home screen, inbox, and profile for latest version of the app

2.1.2. Key Findings

A majority of the issues with the original design had to do with accommodation for smaller devices. A lot of the elements for the prototype (task bar, next page buttons, etc) were designed with bigger phones in mind, and would appear really small on other devices. Additionally, the design language was inconsistent when compared to the SJSU sammy app. A big takeaway from these iterations is the importance of a simple and accommodating UI. This is even more so the case when this app is targeted at College students, who can have a wide range of screen sizes.

2.2. Security & Privacy

2.2.1. Description & Analysis

Due to the nature of the app, sensitive information such as location, email, and passwords are required to be handled. It currently stores usernames, passwords, and the details for listings. These include its title, price, description, image, and general location. To adhere to proper data handling practices, admins of the database do not have access to the passwords. Only the user name is visible for logins. There are intended to be multiple options to quickly log in after first sign in such as a sign in with fingerprint id, but for messages, the usernames of the recipient and sender are encrypted. The admin is only able to view the contents of the message, not the usernames. Timestamp is also provided.

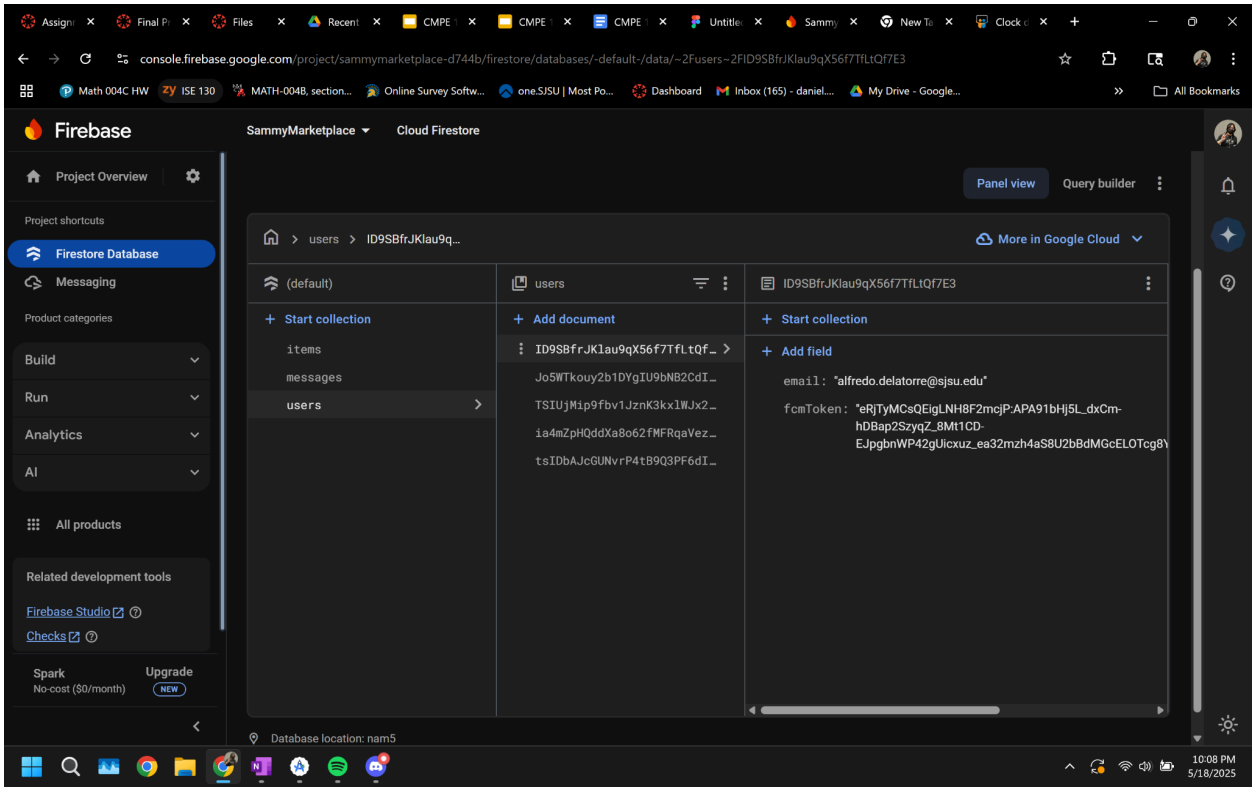


Figure 2.7 - Firebase database view for admin. View of collected usernames example.

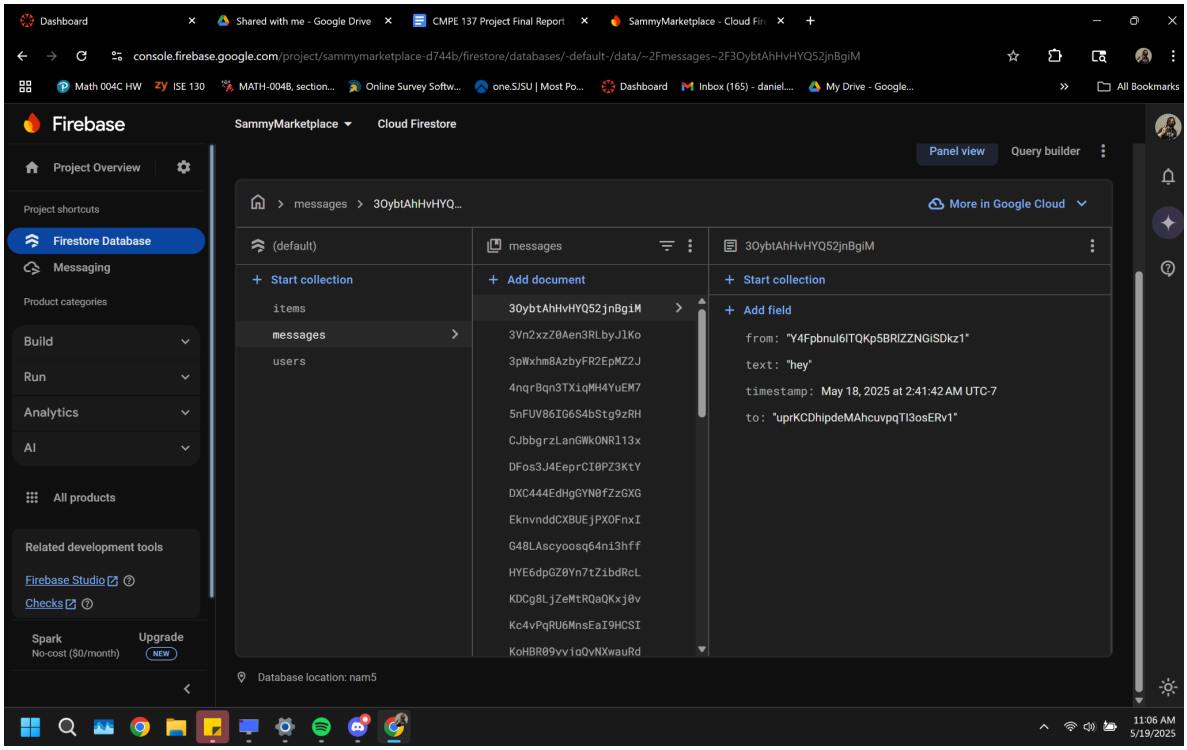


Figure 2.8 - Firebase database view of messages for admin

2.2.2. Key Findings

The importance of proper encryption and data storage became clear with the scope of the project. Because the app enables people to meet in person & requires an SJSU email to sign up, it's important that this sensitive data be properly secured. Firebase makes this process seamless & provides assurance to the end user that even the database admins do not have access to all of the data.

2.3. Connectivity

2.3.1. Description & Analysis

We leveraged **Firebase Realtime Database** for live data exchange and **Cloud Storage** for media uploads. This ensures a seamless experience even with network fluctuations.

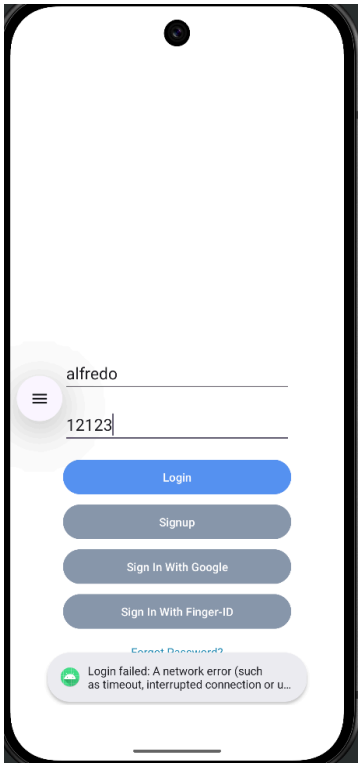


Figure 2.9 -Fail to login due to no internet notification

2.3.2. Key Findings

Implementing Firebase eased the development process heavily, as it made data storage and connection management easy. This experience helped emphasize the need for a proper data storage & connectivity management system.

2.4. Location

2.4.1. Description & Analysis

In the implemented version, precise GPS coordinates are not stored and permission is only asked to allow for future functionality. In future releases, users will be able to optionally add general location data (zip code, city) when posting an item.

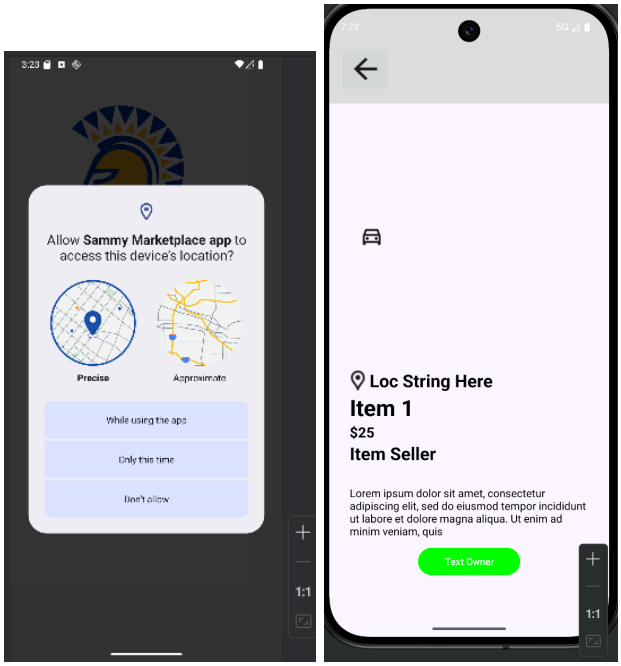


Figure 2.10 & 2.11 -Permission for accessing the device location & product listing screenshot with location

2.4.2. Key Findings

In this use case, location can be used to help students narrow down their search area. Because SJSU is home to a lot of commuter students, many of them may not necessarily be around San Jose all the time. Adding this feature helps let the user know exactly what area they can expect to meet up with the seller in. In terms of other key findings, the implementation of asking the user for permission to use their location helps build trust between the app and the user, as it shows a level of transparency. This implementation is especially important for this app, because it deals with sensitive information and arranges meet up locations.

2.5. Data Management & Exchange

2.5.1. Description & Analysis

Because the app is designed to be a community marketplace, a lot of information has to be stored. It can get a little overwhelming to deal with various types of information, deciding what should remain confidential, and ensuring that the application always has access to this information when called. The implementation of the Firebase Data Storage system streamlines this process and makes it incredibly easy to work with. In its current form, it stores messages, usernames, and item listings. Some information such as passwords and message recipients and senders are kept confidential even to the database admins, but the data is properly encrypted and appropriately managed for sensitive information.

2.5.2. Key Findings

For a project where the stored information is the main focus of the app, the firestore database made the process of storing all sensitive information easy and accessible. This emphasized the importance of having a secure, encrypted database that accurately stores the information. As stated in previous sections, it also only made certain information visible to the database admins, which helps for transparency with the user.

2.6. Notifications

2.6.1. Description & Analysis

In the context of this app, notifications are mainly here to help make the user aware. Case examples include a message from a user expressing interest in an object, any in-app news sent by the admin, or We integrated **Firebase Cloud Messaging (FCM)** to notify users about New Messages/offers, item status updates, and platform announcements. Firebase's documentation made this a fairly straightforward process. All that was required was the device specific token to be able to send messages. Thankfully, there was no payment fee to work with firebase notifications, so its initial implementation went fairly smoothly.

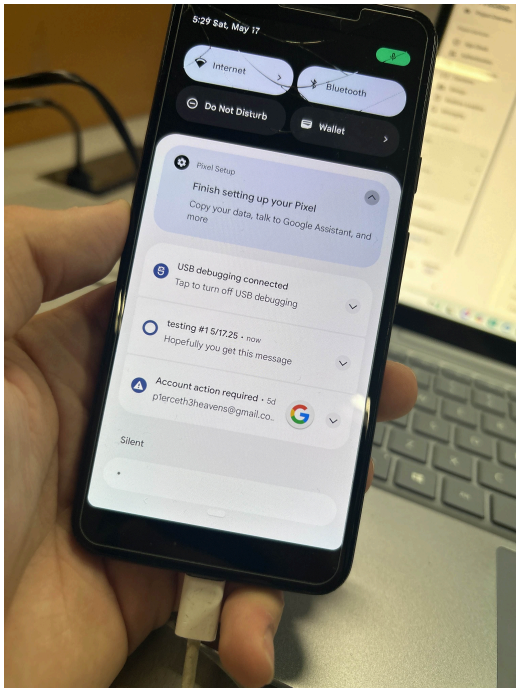


Figure 2.11 - First notification test

2.6.2. Key Findings

A lot of thought has to go into deciding when and what notifications to send to the user. Sending notifications too frequently or ones that contain useless content can turn the user away from using the application. For this use case, it mostly remains for necessary use cases, such as getting messages or important app updates. If properly managed, notifications can promote interaction with the app as well. This balance is important, and firebase makes this easy to consider with its ability to schedule notifications during certain times.

2.7. Evaluation

As the app stands, there's a lot of fine tuning that has to be done. All functions were implemented on their own, but merging the branches together on Github did not wield the expected result. The app is in a state that runs, but has not fully implemented all of the features all at once. For example, connection properly

displays an error when trying to login without wifi, but does not notify the user when they are logged in and lose connection. The code exists for all of the functions in the application, but not all of it translated properly. The layout has changed a few times, but its inspiration from Facebook Marketplace remains. The current state of the layout is mainly functional, but does not have all of the desired design elements from the initial prototype. Firebase implementation went fairly smoothly for the database and notifications, but there were some issues with finding the device token o be able to find notifications. There were a few hiccups with getting the app to run, but it currently is in a state that works.

2.8. Recommendations/Future Work

Should the project continue to improve, the first priority would be touching up the application so all requirements are usable. As said previously, the code is there, but has not translated smoothly for one reason or another. Another recommendation would be to implement more elements from our initial prototype, such as the color scheme and certain layouts like the hamburger menu.

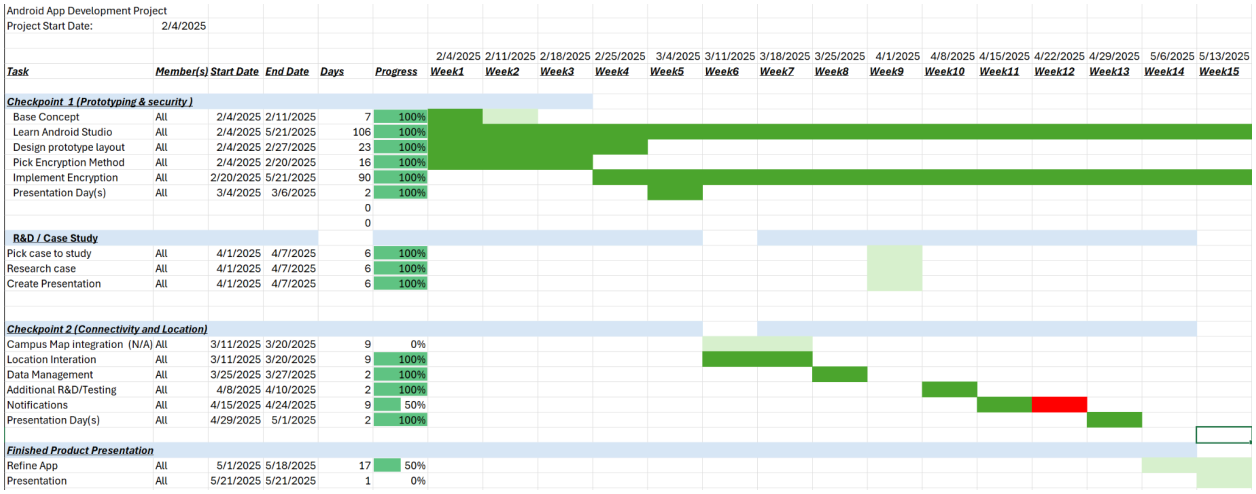
As for future additions to the project, we would like to implement more restrictions on what users can post. We would use OpenAI API to implement that feature where users can't just type and post inappropriate text on their posts.

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3. Project Management

3.1. Final Schedule & Task Distribution (Gantt chart)

Attached below is the Gantt chart for the end of the project. Most goals that were set at the start of the project were completed, but there remains many things to learn to get the app functioning as intended.



Android App Development Project
Project Start Date: 2/4/2025

Task	Member(s)	Start Date	End Date	Days	Progress
Checkpoint 1 (Prototyping & security)					
Base Concept	All	2/4/2025	2/11/2025	7	100%
Learn Android Studio	All	2/4/2025	5/21/2025	106	100%
Design prototype layout	All	2/4/2025	2/27/2025	23	100%
Pick Encryption Method	All	2/4/2025	2/20/2025	16	100%
Implement Encryption	All	2/20/2025	5/21/2025	90	100%
Presentation Day(s)	All	3/4/2025	3/6/2025	2	100%
				0	
				0	
R&D / Case Study					
Pick case to study	All	4/1/2025	4/7/2025	6	100%
Research case	All	4/1/2025	4/7/2025	6	100%
Create Presentation	All	4/1/2025	4/7/2025	6	100%
Checkpoint 2 (Connectivity and Location)					
Campus Map integration (N/A)	All	3/11/2025	3/20/2025	9	0%
Location Interation	All	3/11/2025	3/20/2025	9	100%
Data Management	All	3/25/2025	3/27/2025	2	100%
Additional R&D/Testing	All	4/8/2025	4/10/2025	2	100%
Notifications	All	4/15/2025	4/24/2025	9	50%
Presentation Day(s)	All	4/29/2025	5/1/2025	2	100%
Finished Product Presentation					
Refine App	All	5/1/2025	5/18/2025	17	50%
Presentation	All	5/21/2025	5/21/2025	1	0%

Figure 3.1 - Gantt Chart for the Sammy Marketplace App

3.2. Challenges

Most of the issues can be traced back to having issues translate the figma design over to kotlin in Android studio. There have been 2-3 different drafts for the code translation from figma to kotlin, but most, if not all, have had some weird problems arise as a result. These difficulties set back the development of the app a notable amount. This can be attributed to unfamiliarity with Kotlin and Android Studio. The final product shows a much better understanding of the language, but there remain aspects to be defined. To cite a specific example, making the layouts for the XML file layouts based on figma was a little complicated. Sometimes the layout would be provided by figma and auto translate properly, others would have to be manually recreated in Android Studio, and some would be a mess because of several missing dependencies or unknown drawables.

Another notable challenge with the app initially was with Firebase integration, Setting up the databases for each type of data collected, setting up the Main Activity kotlin file, or properly importing dependencies proved to be a little bit of a challenge. Notifications as well required a device specific token, which took a little bit of work to get from the emulator.

3.3. Learning Outcomes

The Sammy Marketplace project introduced our team to the core tools and workflows of mobile development. Learning Kotlin and working in Android Studio laid a solid coding foundation, while Firebase's cloud-based management, streamlined data storage, user-to-user interactions, and push notifications.

Using Figma for the first time gave us a concrete design roadmap. It gave the team a good understanding of the design process for app development and highlighted the importance of thoughtful UI/UX.

We also dove into often-overlooked details such as notification timing, offline connectivity, and interface responsiveness. Finally, collaborating under an Agile framework provided a real-world glimpse of how distributed teams plan sprints, conduct stand-ups, and iterate quickly. In sum, the project provided an in depth experience into several aspects of mobile software development, ranging from code and design to user experience and team dynamics.

4. Sources

4.1 APA

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